TECHFIX

A WEB-BASED TECHNOLOGY TROUBLESHOOTING AND ACCESSORY COMPATIBILITY ASSISTANT

BSIT Class Project Concept Paper

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TECHFIX: A WEB-BASED TECHNOLOGY TROUBLESHOOTING AND ACCESSORY COMPATIBILITY ASSISTANT

INTRODUCTION

Background of the Project

Technology devices are used for learning, work, communication, and daily tasks. When a phone, laptop, printer, router, or other device develops a problem, users may need to search several sources for troubleshooting steps, product details, compatible accessories, and repair guidance. Information can be scattered or difficult to relate to a particular device. A web-based source that brings these resources together can make basic technology assistance easier to find.

TechFix is a Django-based web application for technology and accessories assistance. Its current prototype brings together a question-and-answer assistant, a product and accessory catalog, accessory-to-product compatibility information, brand pages, and structured repair guides. The assistant matches questions with locally stored catalog and guide information and can use Google Gemini to enrich responses when configured. A local answer path is also available when the external service is not configured or available. The project will further validate these functions and evaluate the system with intended users.

Statement of the Problem and Project Objectives

Users who need technology help may have difficulty locating relevant troubleshooting steps and comparing device or accessory information across separate sources. The project will address this need through a single web application that organizes technology information and provides question-based assistance grounded in the system catalog and repair guides.

The general objective is to develop and evaluate TechFix as a web-based technology troubleshooting and accessories assistance system.

Specifically, the project will:

1. Organize product, accessory, category, brand, and repair-guide information in a searchable catalog.

2. Provide product and accessory details, including compatibility information when it has been recorded.

3. Provide a conversational assistant that can use relevant catalog and repair-guide records to answer technology questions, with optional Gemini-enhanced responses and a local fallback.

4. Provide account functions and user-scoped personal information-record management, together with administrator tools for maintaining catalog data and application users.

5. Evaluate functional correctness and usability through system testing and user acceptance testing with selected intended users.

Significance of the Project

Technology users and students will have a centralized place to find basic troubleshooting guidance and information about devices and accessories. The system is intended to help users identify relevant next steps; it will not replace a qualified repair technician or manufacturer support.

People comparing devices and accessories will be able to review the product information and compatibility notes available in the catalog. Administrators will be able to maintain product, accessory, category, and repair-guide records and manage application users. Signed-in users will be able to maintain their own personal information records. Future student developers may use the project as a reference for combining a Django catalog with locally grounded assistant responses.

Alignment with Institutional Research Agenda and Sustainable Development Goals

The project will support an applied information technology and software development priority by applying web technologies to a practical information-access

problem. The final priority name should be aligned with Kabankalan Catholic College's current approved research agenda.

The project will support Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) by applying digital technology to organize and deliver product and repair information through a web application. This alignment is limited to the project's software-development contribution.

SCOPE AND LIMITATIONS

System Scope and Technical Architecture

The project will cover the TechFix web application and its technology-assistance use case. Its core modules will include the question-and-answer assistant; product, accessory, category, and brand browsing; recorded accessory compatibility; repair guides; user accounts; personal information-record management scoped to the signed-in user; and administrator tools for catalog and user maintenance. Product price, stock, specifications, and compatibility will be shown only when the corresponding records have been entered.

The application will use Python and Django for server-side processing, SQLite as its default relational database, and HTML, CSS, JavaScript, and Bootstrap-based templates for the web interface. Google Gemini will be an optional external service for answer enhancement and will require a configured API key and network access. The local answer path will provide a fallback, but its coverage will depend on the locally available records and matching logic.

Intended users will include visitors seeking basic technology information, registered users accessing account-based features, and administrators maintaining the application. The system will be delivered as a web application and will not include a separate native mobile application.

Limitations

The system will not provide e-commerce checkout, online payment, order fulfillment, or automatic synchronization with retailer inventory. Product and repair

information will depend on records maintained in the application and will not constitute a real-time market feed. Accessory compatibility will be limited to the relationships and notes recorded in the catalog.

Gemini-generated responses may be incomplete or incorrect, and local matching may not find an appropriate guide for every question. The application will not physically inspect devices, confirm a diagnosis, guarantee a repair outcome, or replace manufacturer instructions and qualified repair services. Personal information records will be scoped to the signed-in user, while administrators will be able to access records across accounts. User acceptance testing is planned; no evaluation results are reported in this concept paper.